

Fig 5.4 *Generation AC Waveform*

Now the coil moves to the position 'a' and the angles between the magnetic field and coil is Θ. Then, e.m.f. is equal to BlVsinθ.

Now the coil moves to the position 'b' and the angle between the magnetic field and coil is 90°. Then sin 90 = 1. The emf is maximum. This emf is called as positive maximum.

Now the coil moves to the position 'd' and the angle between the magnetic field and coil is zero. In this position emf is zero.

Now the coil moves to the position 'f' the angle between magnetic field and coil is 90º, sin 90 = 1 and the e.m.f is maximum in magnetic side, so it is called as negative maximum.

Now the conductor moves to the position '0', the emf is zero.

Similarly, the conductor rotates one revolution in the magnetic field. This rotation produces the sine wave form.

a) Cycle

One complete set of positive and negative values of alternating quantity is known as cycle. One complete cycle is said to spread over 360° or 2π radians.

b) Time Period

Time period is denoted by 'T'. The time taken for any wave to complete one full cycle is called the time period.

c) Frequency

The frequency of a wave is defined as the number of cycles that a wave completes in one second. It is denoted by the letter 'F' and its unit is cycles/second or Hertz(Hz). In India, the supply frequency is 50Hz. Frequency is calculated by

$$F = \frac{PN}{120} Hz$$

Where

F = Frequency in Hertz
P = Number of poles
N = Revolution in r.p.m

d) Instantaneous Value

At any given time, it has some instantaneous value. This value is different at different points along the waveform. During the positive and negative cycle, these values are positive and negative respectively.

e) Peak Value

The peak value of the sine wave is the maximum value of the wave during positive half cycle or negative half cycle.

f) Peak Factor

The ratio of maximum value to the r.m.s. value of an alternating quantity is known is peak factor

$$\text{Peak factor} = \frac{\text{Max.Value}}{\text{R.M.S. value}} = 1.414$$

g) Average Value

The average value of the sine wave is the ratio of total area under the half-cycle curve to the distance of the curve

$$\text{Average Value} = \frac{\text{Area under the Curve}}{\text{Base Length}}$$

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